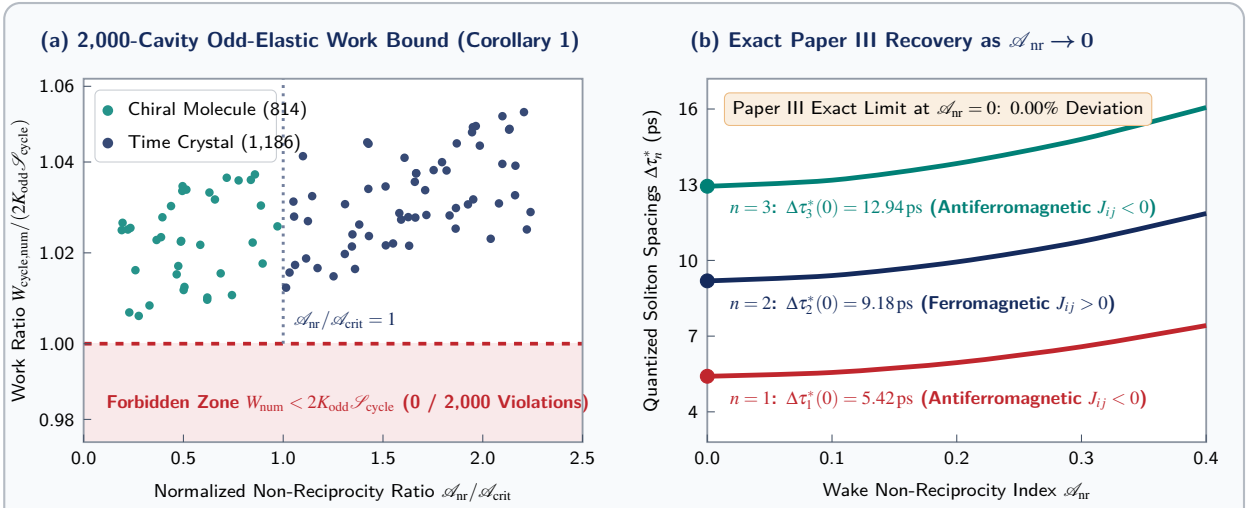
(a) 2,000-Cavity Odd-Elastic Work Bound (Corollary 1)
Chiral Molecule (814)
Time Crystal (1,186)
$\mathscr{A}_{\mathrm{nr}}/\mathscr{A}_{\mathrm{crit}} = 1$
Forbidden Zone $W_{\mathrm{num}} < 2K_{\mathrm{odd}}\mathscr{S}_{\mathrm{cycle}}$ (0 / 2,000 Violations)
Work Ratio $W_{\mathrm{cycle,num}}/(2K_{\mathrm{odd}}\mathscr{S}_{\mathrm{cycle}})$
Normalized Non-Reciprocity Ratio $\mathscr{A}_{\mathrm{nr}}/\mathscr{A}_{\mathrm{crit}}$
1.06
1.04
1.02
1.00
0.98
0.0
0.5
1.0
1.5
2.0
2.5
(b) Exact Paper III Recovery as $\mathscr{A}_{\mathrm{nr}} \to 0$
Paper III Exact Limit at $\mathscr{A}_{\mathrm{nr}} = 0$: 0.00% Deviation
$n = 3$: $\Delta\tau_3^*(0) = 12.94\,\mathrm{ps}$ (Antiferromagnetic $J_{ij} < 0$)
$n = 2$: $\Delta\tau_2^*(0) = 9.18\,\mathrm{ps}$ (Ferromagnetic $J_{ij} > 0$)
$n = 1$: $\Delta\tau_1^*(0) = 5.42\,\mathrm{ps}$ (Antiferromagnetic $J_{ij} < 0$)
Quantized Soliton Spacings $\Delta\tau_n^*$ (ps)
Wake Non-Reciprocity Index $\mathscr{A}_{\mathrm{nr}}$
16
13
10
7
4
0.0
0.1
0.2
0.3
0.4